

Operational freedom in the two-boundary retrocausal framework: an equal-possibility theorem (companion note — hospitality and equality, never metaphysics)

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Metaphysics fence (binding). This note does *not* and *cannot* prove libertarian (metaphysical) free will. What it proves is that a precisely defined *operational* surrogate — the conjunction of four formal properties (equal possibility, non-coercion, openness, efficacy) — holds for every admissible observer of the parent framework, with the same strength for every observer, at the regime- and bundle-scoped grades stated clause by clause. Whether that conjunction deserves the name “free will” is a philosophical judgment outside physics; the note’s own position is compatibilist and is argued, not smuggled. The parent paper defines an observer operationally (ideal-QRF axioms) and explicitly disclaims any appeal to consciousness or agency; this note inherits that disclaimer verbatim. Deliberation, intention, and phenomenology are not modeled anywhere in the framework and are not modeled here.

Abstract

Within the two-boundary retrocausal framework of the parent paper, an observer (“being”) is a worldline–protocol–filtration triple. We define, in the framework’s own vocabulary, four properties that jointly constitute *operational freedom*: (F1) *equal possibility* — at every moment the algebra of available instruments is the same, up to $*$ -isomorphism, for every being; no available option is assigned probability zero by the two-boundary conditioning except through incompatibility with the branch’s terminal record; and every being has the same (dense) operational reach; (F2) *non-coercion* — no axiom determines or probabilistically weights the choice of protocol, no spacelike intervention by any other being can alter the being’s marginal statistics, and the final boundary condition is operationally inert in prior average; (F3) *openness* — the being’s forecast of the terminal record is an unbiased martingale, no variable available before the terminal limit predicts it better, and definiteness arrives only with the records themselves; (F4) *efficacy* — the being’s choices shift the law of the realized branch, lawfully and causally. The main theorem asserts F1–F4 for every admissible being, clause by clause, each at its honest grade and regime: the Part-I spine clauses are theorems of the finite-dimensional core; the field-theoretic clauses carry a named net bundle; the measurement-independence and no-better-predictor clauses are conditional on the parent’s CHT-POVM conjecture plus the thermal-time postulate; and the transfer of the whole conjunction to a single field-theoretic being is scoped by the parent’s type-III lift (Araki branch theorem-grade; sandwich branch conj:tomita). The corollary that gives the note its title: *at any moment, all beings have the same possibilities of choice* — isomorphic choice algebras, a common passive-law terminal distribution, universal reach, common outer time flow (postulate-grade) — while what differs between beings (location, records, thermodynamic state) is exhaustively indexical. An attack ledger and a grade table close the note.

1 Setting and vocabulary

All objects are the parent paper’s. $(\mathcal{M}, g)$ is a fixed globally hyperbolic background with Cauchy temporal function τ and slab $\mathcal{M}_{[\tau_i, \tau_f]}$; ρ_i is the initial state on Σ_{τ_i} and $\{E_k\}_{k \in \mathcal{K}}$ the terminal POVM on Σ_{τ_f} , with backward effects $\Lambda_k(\tau)$ and time-invariant Born weights $p(k) = \text{Tr}(\rho(\tau)\Lambda_k(\tau))$. The

conditioned state is the symmetric-sandwich branch $\rho_2(\tau | k) = \rho(\tau)^{1/2} \Lambda_k(\tau) \rho(\tau)^{1/2} / p(k)$, unitarily covariant with the same generator $H(\tau)$. Local algebras $\mathcal{A}(R)$ satisfy isotony, microcausality (M2) and causal propagation (M3).

Two probability laws must be kept apart throughout, as the parent insists: the *passive* law $\mathbb{P}_\gamma$, defined by construction so that its terminal marginal is $p(k)$ for every protocol, and the *active* sequential-Born law P_M , describing physically performed interventions, whose terminal marginal is the disturbed weight $\tilde{p}(k)$ and need not equal $p(k)$. Statements of “sameness of law” below are passive-law statements; statements of “efficacy” are active-law statements. Conflating the two was the single most frequent defect found by this note’s adversarial review, and the scoping is now explicit at every use.

Regimes. The parent proves its Part-I spine in the finite-dimensional core (all operator identities literal), and its conditioning, Reeh–Schlieder, and no-go–evasion results in the field-theoretic regime (type III₁ net). The type-III lift of the sandwich-state machinery is the parent’s conjecture *conj:tomita*; its Araki-perturbation branch is realised at theorem grade (the parent’s (E4) is closed unconditionally), while the collapse chain transfers type-free through the algebra-free predual theorem. Each clause below names the regime in which its grade is earned; the transfer of the full conjunction to one field-theoretic being is scoped in Remark 2.

Definition 1 (Being; admissibility). *A being is the parent paper’s observer datum: a triple $B = (\gamma, \mathcal{P}_\gamma, \overline{\mathcal{F}}_\tau^\gamma)$ consisting of an inextendible future-directed timelike worldline γ crossing each slice once, a fixed measurement protocol $\mathcal{P}_\gamma = (\{M_j^{(m)}\}_j)_m$ with $M_j^{(m)} \in \mathcal{A}(D_{\varepsilon_m}(\gamma(s_m)))$ for some resolutions $\varepsilon_m > 0$, and the record filtration $\overline{\mathcal{F}}_\tau^\gamma$ it generates under the passive law $\mathbb{P}_\gamma$. A being is admissible for a result if it satisfies the protocol conditions that result assumes: informativeness where terminal record-completeness is invoked, and the record-layer restriction (RP) where the record-sourced geometry is invoked. These are genuine admissibility conditions of the parent, not conveniences of this note. A branch k is admissible if $p(k) > 0$. When frame-invariance of predictions is invoked, B is additionally assumed to satisfy the ideal-QRF axioms of the parent observer appendix.*

Definition 2 (Choice situation). *For a being B , a moment τ , and a resolution $\varepsilon > 0$, the choice set $\mathcal{C}_B(\tau; \varepsilon)$ is the set of all finite Kraus instruments $\{M_j\}$ with $M_j \in \mathcal{A}(D_\varepsilon(\gamma(\tau)))$ and $\sum_j M_j^* M_j = \mathbf{1}$, where $D_\varepsilon(x)$ is the causal diamond of radius ε about x . By construction, every step of an admissible protocol $\mathcal{P}_\gamma$ is an element of the choice set at its moment and resolution.*

Lemma 1 (Functoriality of choice sets). *Any unital $*$ -isomorphism of local algebras carries choice sets bijectively onto choice sets.*

Proof. A unital $*$ -isomorphism preserves adjoints, sums, and the unit, hence the Kraus normalization $\sum_j M_j^* M_j = \mathbf{1}$; its inverse does the same. $\square$

Definition 3 (Operational freedom). *A being B is operationally free if the following four properties hold.*

- (F1)** Equal possibility. (a) For any two beings B, B' , moments τ, τ' , and resolutions $\varepsilon, \varepsilon'$ in the field-theoretic regime, the choice sets $\mathcal{C}_B(\tau; \varepsilon)$ and $\mathcal{C}_{B'}(\tau'; \varepsilon')$ are identified by a $*$ -isomorphism of the underlying local algebras. (b) Non-extinction: for every admissible branch k , no nonzero effect available to B is assigned probability zero by the conditioned state, except on the corner fixed by the branch’s terminal record — a future-boundary datum, block-fixed but in general not yet recorded at τ . (c) Universal reach: from B ’s diamond alone, local operations applied to the conditioned state reach a dense set of global states.

- (F2) Non-coercion. (a) No axiom or equation of the framework determines or probabilistically weights the protocol $\mathcal{P}_\gamma$; the framework's protocol conditions (informativeness, (RP)) delimit which protocols support which of its results, never which protocol the being adopts. (b) Measurement independence holds at the initial slice (the framework is not superdeterministic). (c) No intervention by any spacelike-separated being alters B 's marginal statistics. (d) The final boundary condition is operationally inert for B in prior average: averaged over branches, B 's statistics are exactly those of forward quantum mechanics.
- (F3) Openness. (a) B 's posterior $\pi_\tau^\gamma(k)$ over the terminal record is a martingale on B 's own filtration, unbiased under every record-measurable stopping strategy, and this survives measurement disturbance. (b) No variable available strictly before the terminal limit — and no extension of the framework's statistics beyond $\{E_k\}$ — predicts K better than the posterior. (c) Definiteness arrives only with the records: π^γ collapses to $\mathbf{1}_{\{K=k\}}$ only in the terminal limit, and for horizon-restricted beings may never fully collapse.
- (F4) Efficacy. (a) B 's choice of instrument changes the law of the realized branch: the terminal weights are intervention-dependent under the active law. (b) The consequences are lawful: between interventions, conditioned expectations obey the conserved Ehrenfest dynamics along the realized branch, and the intervention step itself is governed by the instrument update. (c) The consequences are causal, not anticipating: a deferred decision enters the record-sourced geometry only after its record lies in the causal past of the readout.

2 The theorem

Theorem 1 (Operational freedom of every being). *Every admissible being of the parent framework is operationally free in the sense of Definition 3, with the following clause-by-clause provenance, regime, and conditionality.*

- (F1a) ADOPTED + net bundle; field-theoretic regime. *The parent's field-theoretic results assume a Haag–Kastler net of type III_1 on globally hyperbolic spacetime. Granting in addition that every relatively compact diamond algebra is a factor with separable predual and that the net has the split/hyperfiniteness property standard for well-behaved nets, every such algebra is the unique hyperfinite type III_1 factor (Haagerup uniqueness; Connes classification), hence any two beings' diamond algebras — at any two moments, at any two resolutions — are $*$ -isomorphic, and by Lemma 1 their choice sets are identified. Honest fences: (i) factoricity and predual separability are part of the import bundle, not stated at the parent anchor — a nontrivial center at any diamond would break the pairwise identification; (ii) in the finite-dimensional Part I core this clause can fail (unital $*$ -subalgebras of $\mathcal{L}(\mathcal{H})$ at different worldline points need not be isomorphic). Equality of possibility is a genuinely field-theoretic fact, and the type III_1 structure is load-bearing, not decorative.*
- (F1b) THEOREM | net hypotheses; field-theoretic regime; dense-range branch class exact, corner form in general. *The parent's Reeh–Schlieder theorem for ABL conditioning — proved under its stated hypotheses: type III_1 Haag–Kastler net, faithful normal Hadamard state with the Reeh–Schlieder property (ambient imports), $L_k = \Lambda_k(\tau)^{1/2}$ bounded — gives: if L_k has dense range the conditioned state ω_k is faithful and normal on every local algebra, so $\omega_k(F) > 0$ for every nonzero effect F available to any being at any moment. If E_k is projective with nontrivial kernel, faithfulness is preserved exactly on the support corner and lost exactly on $\ker L_k$. The corner is fixed by the terminal record of the branch: a future-boundary datum, block-fixed at*

every τ but in general not yet recorded at τ ; under terminal record-completeness it coincides in the terminal limit with the branch's accumulated record history — the branch's, not the individual being's, since other beings' recorded outcomes are part of it. What the clause rules out is extinction by location, constitution, or by another being's protocol choices; extinction by facts recorded on the branch (by anyone) is exactly what conditioning is.

- (F1c) THEOREM | net hypotheses + DERIVED; field-theoretic regime, dense-range class. Cyclicity of the conditioned vector for every local algebra plus the standard instrument construction (any $A \in \mathcal{A}(D)$ with $\|A\| \leq 1$ is a success branch $M_1 = A$ of the two-outcome instrument $\{A, \sqrt{1 - A^*A}\}$; rescale a cyclic-orbit approximant) yields: for every global normal state φ and every $\delta > 0$ there is an operation in B 's diamond, of strictly positive success probability, whose normalized output state is within δ of φ in vector norm on the GNS space. The derivation from cyclicity is this note's (one step, sketched above); the cyclicity is the parent's. Every being has the same dense reach; postselected, and with possibly minute success probability, but available identically to all.
- (F2a) STRUCTURAL; scoped. By inspection of the axiom set: no axiom or equation determines the protocol or assigns it a probability — the parent introduces $\mathcal{P}_\gamma$ as a fixed non-random external input. What inspection does not yield — and this note's first version wrongly asserted — is that no axiom constrains $\mathcal{P}_\gamma$ at all: terminal record-completeness is a joint identifiability condition on the pair $(\mathcal{P}_\gamma, K)$ that fails for uninformative protocols, and the Part-II record layer imposes (RP), “a genuine restriction on $\mathcal{P}_\gamma$, not bookkeeping.” These are admissibility conditions delimiting which protocols support collapse and record-sourcing; they do not weight, forbid, or select the being's choice among admissible protocols. Hospitality, correctly scoped: the framework has no resource to force or bias the choice; it does say which choices yield records that identify the outcome and may source geometry.
- (F2b) THEOREM | Conj. CHT-POVM + Pos. thermal-time. The parent's combined Bell-style evasion theorem — whose standing hypotheses are the CHT-POVM conjecture, the thermal-time postulate, terminal record-completeness, and bulk no-signalling — places the framework as “non-dynamical (all-at-once), future-input-dependent retrocausal without operational signalling” and states verbatim that it is not superdeterministic: measurement independence at the initial slice is retained; what is abandoned is No-Retrocausality, and the resulting future-input dependence of past hidden variables is a modular invariant, not a tunable parameter. The parent's own concessions are inherited unshrunk: the evasion is of causal fine-tuning, not Catani–Leifer operational fine-tuning (“lawful fine-tuning, not its absence”), and a general adaptive-interventions analysis is an open scope item.
- (F2c) THEOREM; Part-I core (finite-dimensional). Bulk no-signalling: for spacelike separated regions, the choice of instrument in one leaves the outcome marginals in the other exactly invariant; the past-to-future direction uses microcausality and causal propagation, the future-to-past direction not even that. Scope, stated plainly: this protects B 's k -summed marginals. It does not protect per-branch post-selected statistics, and it does not make the shared terminal law insensitive to others (see Remark 4).
- (F2d) THEOREM in the Part-I core; in the field-theoretic regime via the Araki-conditioning branch, with the sandwich-branch lift carried by conj:tomita. The prior-average identity $\mathbb{E}_{\mathbb{P}_\gamma}[\rho_{\text{RC}}^\gamma(\tau)] = \rho(\tau)$ and the reduction corollary: averaged over branches, every expectation any being can form is the ordinary forward-QM expectation from ρ_i . The future boundary shapes which past is realised and provably cannot be used to send a message into it.

- (F3a) THEOREM in the Part-I core; type-free in general (disturbance-robust) + DERIVED. The posterior is a martingale on the record filtration by the tower property; the martingale property survives arbitrary measurement disturbance — disturbance changes the measure, not the Bayesian structure — and the collapse chain is algebra-free (valid verbatim for type III preduals). Derived here (standard probability, one step): for every record-measurable stopping time T , $\mathbb{E}[\pi_T^\gamma(k)] = \pi_0^\gamma(k)$ (optional sampling for a bounded martingale closed by $\mathbf{1}_{\{K=k\}}$). This is unbiasedness under stopping strategies, not non-improvement of accuracy: expected forecast accuracy does improve as records accumulate — that is exactly F3c’s collapse — but no strategy can bias the forecast or extract an expectation advantage over the current posterior.
- (F3b) THEOREM | Conj. CHT-POVM + Pos. thermal-time. The Colbeck–Renner clause: the terminal PVM is the maximal abelian centraliser of the tracial state on the cosmological-horizon algebra, so no predictive extension beyond $\{E_k\}$ exists. Stated with the correct time index: no variable available strictly before the terminal limit, and no extension of the framework’s statistics, predicts K better than the posterior. (Under terminal record-completeness K is a.s. a function of the complete pre-terminal record history — this note’s first version wrongly denied it; openness is openness-at-every-interior-moment, not ontological chanciness of the completed block. The compatibilist reading of this fact is discussed, not asserted, in Section 4.)
- (F3c) THEOREM under terminal record-completeness (Part-I core; type-free via the predual theorem); quantitative residual form without it. Collapse $\pi_{s_n}^\gamma(k) \rightarrow \mathbf{1}_{\{K=k\}}$ holds only in the terminal limit, and for accessible sub-filtrations the residual entropy $H(K | \mathcal{G}) \geq 0$ can remain strictly positive: definiteness never precedes the records.
- (F4a) THEOREM; Part-I core, active law. The Born weight is not invariant under bulk interventions: under the active law the terminal marginal is the disturbed weight, dependent on the chosen instruments; the change is a causal influence on a non-local quantity, not a signal, and backward influence is permitted within the post-selected sub-ensemble.
- (F4b) THEOREM; Part-I core only (the Ehrenfest identity is formulated by the parent in the finite-dimensional setting and no field-theoretic analogue is claimed here). Outcome-wise conservation of conditioned expectations and the conditioned Ehrenfest theorem: between interventions, conditioned expectations along the realized branch obey the ordinary conserved Hamiltonian dynamics; the intervention step itself is the instrument update.
- (F4c) STRUCTURAL | record layer (RP), per worldline. The parent’s delayed-choice scope statement holds by construction of the (RP)-admissible protocol class — a deferred commit/erase decider is outside the class until its decision becomes a record in the causal past of the readout — supported at proposition grade by protocol-counterfactual independence of the metric outside record cones. Open corner, inherited unshrunk: the reconciliation of distinct worldlines whose deferred choices are mutually spacelike is the parent’s open multi-worldline corner; the multi-being form of this clause is not claimed.

Remark 1 (Proof discipline). Each clause is one of: quoted from a parent theorem at its printed grade (with label and line anchor in the source comment); imported from the ambient literature at ADOPTED grade; verified by inspection of the parent axiom set (STRUCTURAL); or a one-step corollary derived in this note with its proof sketched in place (DERIVED: the instrument construction in F1c, the optional sampling step in F3a). No clause is asserted above the grade of its weakest ingredient; the conjunction is graded in Table 1.

Remark 2 (Regime scoping of the conjunction). *No single regime carries all thirteen clauses at unconditional grade, and the theorem does not pretend otherwise. In the finite-dimensional Part-I core, the spine $F2a$, $F2c$, $F2d$, $F3a$, $F3c$, $F4a$, $F4b$ is theorem-grade, but $F1a$ can fail and $F1b/F1c$ are not available (no Reeh–Schlieder). In the field-theoretic regime, $F1a$ – $F1c$ hold under the net bundle, $F3a/F3c$ transfer type-free via the parent’s algebra-free predual collapse theorem, two-boundary conditioning itself is theorem-grade on the Araki branch (the parent’s $(E4)$, closed unconditionally), and the sandwich-state form of $F2d$ rides on **conj:tomita**, which is hereby named in the bundle rather than silently consumed. $F4b$ is claimed in the core only. A single field-theoretic being therefore satisfies the conjunction at the grades: $F1$ as printed; $F2a$ structural; $F2b$ conditional; $F2c$ in the core idealization; $F2d$ on the Araki branch; $F3$ as printed; $F4a$ – $F4b$ in the core idealization; $F4c$ per worldline. This is the strongest honest form of the claim; concentration, not concealment.*

3 The equality corollary

Corollary 1 (Equal possibility of all beings). *At any moment, all beings have the same possibilities of choice, in the following precise and graded sense.*

- (i) Same options (grade: $F1a$ ’s). *Their choice sets are identified by $*$ -isomorphism of their diamond algebras: there is no algebraic fact about what one being could do that is not equally a fact about every other.*
- (ii) Same passive law (grade: THEOREM for worldline-independence; ADOPTED|Pos. thermal-time bundle for frame-independence). *Under the passive two-boundary law the terminal distribution $p(k)$ is worldline-independent, and, for ideal QRF observers, frame-independent — the frame-independence half resting on the parent’s observer-invariance proposition, itself an adoption conditional on the thermal-time postulate and the gravity-sector realisation, and weakening for non-ideal QRFs. Under the active law the realized terminal marginal is protocol-dependent — that is $F4a$ ’s efficacy, shared symmetrically by all beings, not an inequality between them.*
- (iii) Same reach (grade: $F1c$ ’s). *Each being’s local operations reach a dense set of global conditioned states; none is granted a region of state space inaccessible to another.*
- (iv) Same openness (grade: $F3a$ unconditional; $F3b$ conditional). *Each being’s forecast of the realized branch is bounded by the same martingale structure, and no being possesses — or, conditional on the $F3b$ bundle, could possess — a better predictor. No being is the universe’s oracle.*
- (v) Same time, modulo gauge (grade: postulate-conditional — Pos. thermal-time, with the parent’s co-scoping residual and **conj:tomita** seed behind it; not theorem-grade). *Thermal time is state-dependent only at the inner level; the Connes cocycle is inner, so beings share a canonical outer flow, their clocks differing exactly by their thermodynamic state.*

What distinguishes beings is exhaustively indexical: which subalgebra (location), which filtration (history), which inner cocycle (thermodynamic state), and — per $(F1b)$ — which options the branch’s terminal record forecloses. None of these alters the abstract possibility structure, the passive law, the reach, or the openness. The weakest-link rule of Table 1 covers every anchor cited by an item, parent postulates and adopted propositions included, not only F -clauses.

Remark 3 (What equality does not say). *Equality of possibility is not equality of knowledge or of practical capacity. Horizon-restricted beings carry strictly positive residual entropy about the terminal record while others may not; and the magnitude of a being’s influence on the realized law depends on its location and instruments. The corollary asserts equality of the possibility structure, passive law, reach, and openness — not of epistemic access, resources, or leverage. Conflating these would overclaim, and the parent’s own horizon-entropy corollary refutes the conflation.*

Remark 4 (Equality and mutual influence). *Non-coercion is a statement about marginals and must not be inflated. What is theorem-protected (F2c) is each being’s k -summed spacelike marginal statistics. What is not protected — and is in fact asserted by F4a, symmetrically for every being — is the shared terminal law and the per-branch post-selected statistics: any being’s intervention shifts the disturbed weights $\tilde{p}(k)$ that all beings jointly face, and backward influence within a post-selected sub-ensemble is permitted by the parent. In a multi-agent world, beings are mutually influential at the level of the branch law while mutually non-coercive at the level of marginals — the same distinction the parent draws between causal influence and signalling. Freedom here is freedom among interdependent beings, not isolation.*

Remark 5 (Why this is freedom-shaped and not merely randomness). *Chance alone is not freedom; the conjunction is what carries the content. (F2) says nothing outside the being forces or biases its options at the level of its marginal statistics — not the initial data (measurement independence), not other beings (no-signalling), not the future boundary (prior-average inertness) — while Remark 4 states exactly the sense in which beings remain mutually influential; (F3) says nothing, inside or outside the framework, forecasts the resolution better than the being’s own posterior; (F1) says the option set is never degraded by the conditioning except through the branch’s terminal record; (F4) says the choice genuinely reshapes the law of what is realized, including retro-shaping the post-selected past, lawfully and causally. A fixed future boundary that is provably non-coercive at the marginal level, non-predictive, option-preserving, and choice-sensitive is the strongest freedom-compatible structure a two-boundary theory can offer.*

4 What is not proven

- **CONCEDED** *Libertarian free will.* Per realization, the two-boundary block — initial data, terminal record K , and the record history — is a fixed fact, and under terminal record-completeness K is a.s. determined by the complete pre-terminal record history. The framework offers no ontological “could have done otherwise” over and above (F2a)’s protocol-externality and (F3b)’s no-better-predictor clause. The theorem is compatibilist in kind: it proves the fixity is undetectable in prior average (F2d), non-coercive at the marginal level (F2), non-predictive from any interior moment (F3), and option-preserving (F1). It does not dissolve the block; it proves the block cannot be used against its inhabitants.
- **CONCEDED** *A theory of deliberation.* The protocol $\mathcal{P}_\gamma$ is an unanalyzed input. The framework does not model how a being chooses; it proves the choice, however made, enters as an external variable the dynamics does not weight (F2a) and cannot anticipate (F3, F4c). This is a hospitality theorem, not a mechanism of agency.
- **CONCEDED** *Unconditional status of every clause.* (F2b) and (F3b) ride on Conj. CHT-POVM and Pos. thermal-time — two named points of conditionality, the first being the same one the parent flags for its no-go evasions; (F1a–F1c) carry the net bundle (III₁, Hadamard/RS state, factoriality, separable predual, split); the field-theoretic sandwich form of (F2d) rides

on conj:tomita; (F4b) is core-only; (F4c) is per-worldline with the multi-worldline corner open. Failure of CHT-POVM degrades (F2b) to the parent’s symmetry-natural fine-tuning fallback but eliminates (F3b)’s no-predictive-extension clause outright — no fallback for it exists in the parent — leaving openness supported only by the unconditional martingale and collapse-timing clauses (F3a, F3c).

- **CONCEDED** *Derivation of the Born weight.* The equality-of-law clause inherits the parent’s status register: the Born weight is an input, not a derived quantity.

5 Attack ledger

Anticipated referee attacks, recorded in the ledger idiom; each with a verdict. FW-A1 through FW-A7 were drafted with the note; FW-A8 and FW-A9 were added after the adversarial swarm run, which also forced the re-tensing of FW-A3.

- FW-A1** *“Block fatalism: K is fixed, so nothing is free.”* Verdict: **CONTAINED**, with grades displayed. The fixity is per-realization and per-protocol: the joint law is a function of the being’s free protocol choice (F4a, unconditional), the boundary’s fixity is provably inert in prior average (F2d, unconditional in the core), and no interior-moment variable predicts the resolution (F3a unconditional; the no-extension leg F3b conditional on CHT-POVM+thermal-time). The residue — “given the whole block, this block” — is a tautology, conceded in Section 4 and carrying no operational content by (F2d).
- FW-A2** *“Equal possibility is vacuous algebra.”* Verdict: **REFUTED**. The isomorphism clause fails in the finite-dimensional core and holds precisely because local algebras in the field-theoretic regime are the unique hyperfinite III_1 factor. Equality of beings is a contingent, load-bearing structural fact about the universe’s local algebras, not a notational triviality.
- FW-A3** *“Projective terminal effects extinguish options, and CHT-POVM selects a PVM.”* Verdict: **CONCEDED-AND-SCOPED**, re-tensed after review. On projective branches faithfulness holds on the support corner and fails exactly on $\ker L_k$ (F1b). The extinguished options are those incompatible with the branch’s *terminal record* — a future-boundary datum, not the being’s past. The defence is therefore not the past-constraint trope (this note’s first version used it; the swarm killed it) but non-coercion: the future-record constraint is block-level conditioning that is operationally inert in prior average (F2d), undetectable and unusable from inside, and identical in kind for every being. Constraint by the branch one is in fact on is what conditioning *means*; what freedom requires — and what the theorem delivers — is that the constraint is not wielded by anything against anyone. Cross-reference FW-A1.
- FW-A4** *“Retrocausal setting-dependence is superdeterminism by the back door.”* Verdict: **REFUTED AT GRADE**. The dependence runs from settings to past hidden variables within post-selected sub-ensembles — not from initial data to settings. Measurement independence at the initial slice is retained verbatim in the parent; cosmic Bell tests are structurally blind to future-boundary conditioning. Grade honestly carried: conditional on CHT-POVM+thermal-time, with the parent’s lawful-fine-tuning concession inherited.
- FW-A5** *“The ‘beings’ are QRF triples; no agent, no will.”* Verdict: **CONCEDED BY DESIGN**. See the metaphysics fence. The theorem’s content survives the deflation: whatever does the choosing, the framework does not weight it, cannot coerce it at the marginal level, cannot foresee it, and does not asymmetrically endow it.

- FW-A6** “*Equality is false: a being behind a horizon knows less.*” Verdict: CONCEDED AND QUARANTINED. Epistemic access is provably unequal (horizon-hidden entropy); the corollary claims equality of possibility structure, passive law, reach, and openness only (Remark 3).
- FW-A7** “*Universal reach is postselection theater: the success probability can be astronomically small.*” Verdict: CONCEDED-AND-SCOPED. (F1c) is a possibility statement, graded as such; it asserts the same dense reach for all beings, not cheap reach for any. The note’s freedom notion nowhere trades on confusing possibility with facility.
- FW-A8** “*Beings coerce each other through the shared law: A’s intervention changes the $\tilde{p}(k)$ that B faces.*” Verdict: CONCEDED-AND-SCOPED (Remark 4). True, symmetric, and not coercion in the theorem’s sense: B’s spacelike marginals are theorem-protected (F2c); the shared branch law is influence-sensitive for everyone alike (F4a). Mutual influence at the law level with mutual non-coercion at the marginal level is the framework’s precise shape of interdependent freedom; the note claims nothing stronger.
- FW-A9** “*Agents reasoning about agents: Wigner’s friend and Frauchiger–Renner break shared-world reasoning.*” Verdict: DELEGATED AT GRADE. The parent’s Bell-evasion theorem, item (3), makes nested agents’ reasoning chains well-defined relative to the unique terminal record, blocking the Frauchiger–Renner contradiction — THEOREM|CHT-POVM+thermal-time, the same bundle as F2b/F3b. A being coherently manipulated as another’s measured system is outside the record-generating protocol class; the equality corollary speaks of record-generating beings.

6 Grade table

Clause	Content	Grade	Regime	Parent anchor
F1a	isomorphic choice algebras	ADOPTED + net bundle	QFT	RS-ABL hyp.; ambient
F1b	non-extinction of options	THEOREM net (dense-range; corner o.w.)	QFT	RS-ABL (i)–(iv)
F1c	universal reach	THEOREM net + DERIVED	QFT	RS-ABL (ii)
F2a	no weighting of protocol	STRUCTURAL (scoped; adm. conditions exist)	both	filtration section
F2b	measurement independence	THEOREM CHT-POVM+th.-time	QFT	Bell evasion (1)
F2c	marginal non-coercion	THEOREM	core	bulk no-signalling
F2d	inert future boundary	THEOREM core; conj:tomita QFT	both	prior-average identity
F3a	martingale openness	THEOREM + DERIVED; type-free	both	posterior lemma; predual thm.
F3b	no better predictor	THEOREM CHT-POVM+th.-time	QFT	Bell evasion (4)
F3c	definiteness only with records	THEOREM record-completeness; type-free	both	collapse thm.; horizon cor.
F4a	efficacy on the branch law	THEOREM (active law)	core	$p(k)$ non-invariance
F4b	lawful consequences	THEOREM	core only	conservation; Ehrenfest
F4c	causal, not anticipating	STRUCTURAL (RP); per worldline	core	delayed-choice scope
Cor. 1	equality of beings	weakest-link over <i>all</i> cited anchors	—	incl. Pos. th.-time, obs.-inv.

Table 1: Grades carried faithfully; the corollary’s weakest-link rule ranges over every anchor its items cite — F-clauses, parent postulates, and adopted propositions alike. Unconditional spine (each in its stated regime): F2a (scoped), F2c, F2d (core), F3a, F3c (|record-completeness), F4a, F4b (core). F1b/F1c are theorem-grade under the net bundle; nothing in this note is unconditional in a regime it is not stated for.

Ambient references (import list)

Cited from standard knowledge; primary-source verification pending per house rule before any import into the parent.

- U. Haagerup, *Connes' bicentralizer problem and uniqueness of the injective factor of type III_1* , Acta Math. 158 (1987) 95–148. [F1a]
- A. Connes, *Classification of injective factors*, Ann. Math. 104 (1976) 73–115. [F1a]
- D. Buchholz, C. D'Antoni, K. Fredenhagen, *The universal structure of local algebras*, Commun. Math. Phys. 111 (1987) 123–135. [F1a: split, and the hyperfinite- III_1 -factor-up-to-center reduction that makes the factoriality hypothesis explicit]
- C. List, *Free will, determinism, and the possibility of doing otherwise*, Noûs 48 (2014) 156–178. [compatibilist framing, fence]